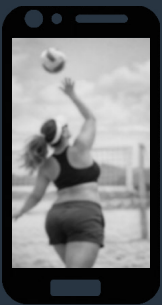
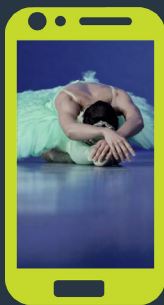
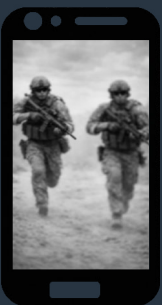
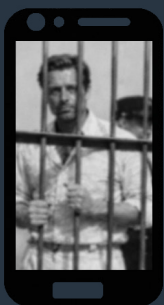
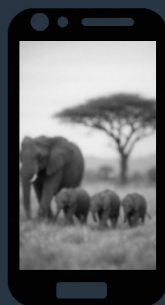
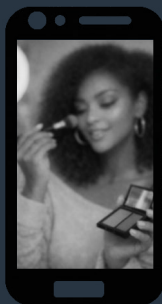
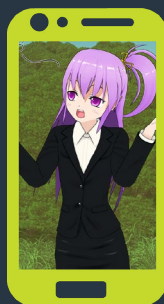
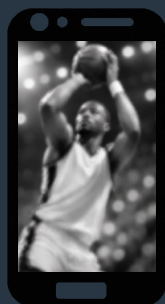


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



JUSTIN GRANDINETTI
CHARLES ECENBARGER

MOBILE PERSONALIZATION

*The History and Theory of
Mobile Streaming Media*

by
Justin Grandinetti &
Charles Ecenbarger

Mobile Personalization: The History and Theory of Mobile Streaming Media

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CHAPTER FIVE

The Future of Mobile Streaming

Everybody's on a barge floating down the endless stream of great TV.

—The Arctic Monkeys

Attempting to predict the future is an exercise that often ages poorly. Tech CEOs see stocks benefit from bombastic hysteria and editorials gain traffic from hot takes. But academic work should tread more carefully. We previously mentioned George du Maurier's 1897 comic depicting an imagined Edison invention called the telephonoscope, where two individuals can view, listen to, and interact with a public setting from the comfort of their own home. This playful fiction certainly seems prescient of a certain type of streaming now well over a hundred years later. But for every technological prognostication that comes true, there are countless others that miss the mark. Science fiction like *Blade Runner* had audiences thinking that synthetic human replicants would be born in 2016. *2001: A Space Odyssey* showed humans traveling to Jupiter, when manned spacecraft have not ventured past the moon now some twenty years later than the movie's setting. Even *Neuromancer*, a sci-fi novel lauded for its prescience on the internet and AI, still imagined characters going through their lives in ways that echoed the 1980s context in which it was written; no one is glued to a screen in their pocket.

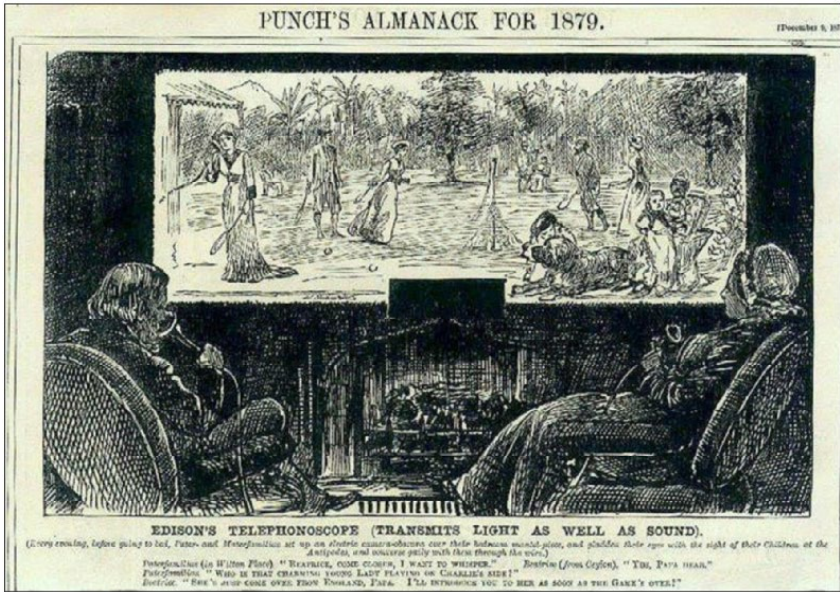


Figure 5.1. George du Maurier's telephonoscope comic, 1897. Uploaded to ResearchGate by Benjamin Burroughs.

Science fiction might seem like an easy target to pick on (lovingly, of course), but even more impactful and bombastic prognostications are seen time and time again in contemporary pitches by big tech. Tesla has promised the imminent advent of self-driving cars for almost a decade.¹ Companies pitching the future of AI and the accompanying global disruptions seem to be more vaporware than actuality.² Hype has become critical to courting tech investors, even if the claims are often dubious.³ In short, it's probably safer to assume big tech has it wrong than the other way around.

Streaming media is no exception when it comes to slightly wrong and even outright incorrect predictions. In 2014, for example, Netflix CEO Reed Hastings correctly asserted that Nielsen ratings could not capture accurate audience numbers for Netflix programming because of the amount of phone and iPad views.⁴ While he was correct about the growing number of mobile views, he also predicted the fairly imminent death of traditional television,

¹ Trisha Thadani and Rachel Lerman, "7 Times Elon Musk's Predictions About Self-Driving Cars Fell Short," *The Washington Post*, October 10, 2024.

² Emily Stewart, "We're Entering the Age of AI BS," *Business Insider*, March 27, 2024.

³ Gemma Milne, *Smoke & Mirrors: How Hype Obscures the Future and How to See Past It* (Robinson, 2020), 3.

⁴ John Hect, "Netflix Chief Downplays Nielsen Plans to Measure Streaming Service Viewership," *Hollywood Reporter*, November 24, 2014.

declaring, “It’s kind of like the horse, you know, the horse was good until we had the car . . . the age of broadcast TV will probably last until 2030.”⁵ It is true that a majority of US households now do not have cable TV, and that “cord cutting” is increasing yearly.⁶ In 2021, 69 percent of surveyed respondents stated that the high cost of cable and satellite was the driving factor to their switch to streaming-only.⁷ Like many other sociotechnical disruptions, however, what started out with utopic promise has become more of a mixed bag. The freedom of cord-cutting cable services for one cheap streaming alternative has come to look a lot more like cable than ever before. The promise of liberty from inconveniences such as advertisements and time-specific content has also come crashing back to earth. And even when it comes to user-generated content streaming platforms, the social optimism of Web 2.0 has given way to the dangers of mis/disinformation and mental health issues.

As noted throughout book, delineating a singular origin date for streaming media is challenging. In a popular sense, the streaming revolution for audio-visual content began to take off in 2005 and ensuing years, which means we have been living with streaming as a normal part of everyday life for roughly two decades. All of this is to say that in a chapter about the future of mobile streaming, we are wary of going too far out on a limb. Chances are that streaming in all its forms is here to stay. Individuals—and particularly younger audiences—seem to prefer mobile and short-form content. But that doesn’t mean that cinematic and televisual-style content is not long for this world. If there is any trend we can safely assume will continue, it’s that individuals will keep streaming.

Up until this not-so-bold claim, we’ve spent this book looking to the past and present of mobile streaming for what we believe to be good reason. Our project is inspired by media archaeological accounts, particularly those attentive to understudied portable and mobile technologies. We eschew the search for a “eureka moment”—a simple narrative of a lone inventor changing the world. Instead, we’ve attempted to show the complex lineage of mobile streaming, as well as the complex mix of technological changes and social imaginings that underpin our contemporary practices. We did so by looking at material changes in technologies—not just the screen of

⁵ Hect, “Netflix Chief Downplays Nielsen Plans.”

⁶ David Satin, “Report: 52% of US Households Do Not Have Cable; Nearly 70% of Streaming Users Plan to Cycle,” *The Streamable*, February 13, 2023.

⁷ Lee Rainie, “Cable and Satellite TV Use Has Dropped Dramatically in the US Since 2015,” Pew Research Center, March 17, 2021.

the smartphone, but also the infrastructures, protocols, wireless networks, APIs, and platforms that make streaming anytime and anywhere possible. We've also highlighted changes in the discursive—that is, how powerful forces shape the idea of mobile streaming. Rather than merely accepting that mobile streaming is a relatively new idea, we've traced the emergence of a desire for a sociotechnical orientation to time and space as something where individuals could access information and entertainment, while corporate interests could extract value. We call this lineage mobile personalization.

We began chapter 1 by tracing the historical roots of mobile personalization. Streaming media is inexorably linked with the history of the televisual and cinematic—something that's reinforced through statements by representatives of major streaming platforms. Our focus was not to outright reject the relevance of the film and television to contemporary mobile streaming, but to instead complicate these narratives. Cinema and later television were critical technologies to the temporal and spatial reorganization of industrial modernity and mass society. But alongside the fixed, brick-and-mortar structures of the cinema and the domestic site of emergent televisual consumption by the nuclear family were imaginings of these technologies intertwined with the mobile and portable. A media archaeological approach highlights the drive-in cinema and the portable television as oft-ignored technologies on the margins of commonly accepted historical narratives. Most critically, these mobile and portable takes on the televisual and cinematic demonstrate that the desire to expand spaces of entertainment and information while also monetizing spaces through novel technologies existed long before the smartphone.

Chapter 2 considered the hidden infrastructures and platform systems that make mobile streaming possible. What seems effortless—a video on your phone in an airport line—is actually the product of a vast, layered architecture of networks, cloud systems, platforms, and data pipelines. Mobile streaming depends not just on smartphones or apps, but also on global telecommunications networks, data centers, algorithmic curation, and platform business models built to extract data and monetize attention. We showed how platforms like Netflix, YouTube, Spotify, TikTok, and Twitch operate as both services and infrastructures, using APIs, wireless networks, and big data to shape media access, recommendation, and consumption. These architectures of mobility have shifted media from static, scheduled delivery to a personalized, always-on, real-time system deeply entangled with surveillance and commercial interests. Power is increasingly concentrated

in a few platform companies that control both the means of delivery and the flow of information. This convergence reshapes culture, narrows choices through algorithmic curation, and extends corporate influence into daily life. Recognizing how these systems operate, and who benefits, is essential for understanding the stakes of the mobile streaming era. Streaming media is not just about content; it is about control, infrastructure, and the future of digital culture.

Chapter 3 continued this trajectory through an examination of how mobile streaming has reshaped everyday life and redefined media industries. Platforms like TikTok, YouTube, Twitch, Instagram, and Apple Music have built seamless, always-on ecosystems where content and user data move fluidly across devices. This convenience, however, is engineered to serve deeper commercial goals such as maximizing user engagement, extracting data, and monetizing every interaction through advertising, subscriptions, microtransactions, and platform integration. Platform mobility and cloud infrastructure work together to tether users to corporate ecosystems, blurring the line between leisure and labor. Mobile streaming has become embedded not just through technology, but also through strategic design choices that normalize continuous connection and consumption. These platforms are not neutral; they shape behaviors, speech, and cultural norms based on business imperatives. The global reach of mobile streaming also highlights inequalities in infrastructure, regulation, and platform power. As mobile streaming becomes the dominant mode of media consumption, the challenge ahead is to ensure that these powerful systems serve broader societal needs, not just corporate profit. Understanding the commercial and political architecture behind mobile streaming is essential if we hope to navigate, challenge, and reshape the digital environments we increasingly live within.

In the final chapter, we considered how the move to mobile streaming complicates some of the key academic literature on audio-visual media. Specifically, mobile streaming necessitates reconsideration of elements like flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data. These concepts originated in scholarship about audio-visual media, including more recent accounts of streaming. However, mobile streaming remediates some of these academic understandings. Flow, the sequence of programming that creates an unbroken experience for audiences, is remade in the era of the endless video scroll. Liveness and sharedness, aspects of mass media that seemed endangered by early streaming, are made immensely relevant to mobile streaming via social connectivity. Mobile

streaming practices also foster new relationships to screens and space, as well as mobile and hybrid engagement practices. And finally, the elements of algorithmic suggestion and big data collection, storage, and processing are shaped by mobile devices and location. The goal, of course, is to consider what it means to “stream” in a mobile world.

Taken together, we’ve spent a lot of this book looking backward to better understand the present moment of mobile streaming. Just a few short years ago, a platform like TikTok was an unrealized idea. Now, both conventional streaming platforms as well as rival social media platforms chase TikTok’s runaway success. Short-form streaming was not realized through the corporate releases of Quibi, but instead through social media content. At the same time, audiences watch a host of other content via mobile devices, be it shows, parts of movies, advertisements, game streams, adult content, news, and advertisements. It’s almost become redundant to say “mobile streaming,” considering just how much streaming engagement now comes via mobile devices.

The plurality of mobile streaming practices aligns with our material-discursive archaeological approach. That is, we aim to decenter one simple narrative of mobile streaming, as well as to avoid deterministic thinking. We aren’t sure where streaming goes next. And it’s difficult to decipher whether what’s next will be “good” or “bad.” Social media streaming allows people to come together, to share information and be entertained. It also allows misinformation and disinformation to circulate persuasively and effectively. Streaming platforms increasingly optimize for mobile, but as noted throughout the book, these same platforms are data hungry and engage in a type of surveillance and subtle manipulation. Not to mention the fact that the production of content has gone from taking risks to data-based in a way that can often feel soulless. As examined in chapter 3, the layers of technologies that underpin streaming are subject to corporate consolidation. Just what happens when major forms of communication are controlled by a small number of actors is a question with some serious consequences. And how consolidation plays out across geopolitical borders, such as the example of multiple countries banning TikTok, remains to be seen.

Mobile streaming is no longer novel but a seamless part of daily life, threading through our routines in ways we barely notice. We carry it with us from morning commutes to late-night wind-downs, expecting entertainment anywhere. That ubiquity is a mixed blessing. It can enrich moments with music or video, but it can also fragment our attention, shape our habits,

and quietly trade away privacy and time. Streaming services now offer both promise and puzzle, connection and distraction, all woven together in the background of our days.

Taken together, while we might not know exactly what streaming will look like five, ten, or twenty years from now, we can speak to a few certainties. Streaming is likely to continue to change, intermix with new contexts, and further integrate into media ecosystems. If previous trends hold true, the distinction between mobile and fixed is likely to become blurrier and perhaps less meaningful. Most of the world, particularly younger individuals and those in the Global South, are far more likely to access the web via mobile devices. For them, mobile is the norm. Short-form mobile streaming, at first something trendy, hip, and cutting edge via vanguard platforms like TikTok, is now an integral part of more legacy platforms like Facebook, which largely trends toward older users. In essence, what mobile streaming looks like will change and further integrate into everyday contexts. However, the perspective of mobile personalization allows us to understand that these changes go beyond the merely technological and are instead sociotechnical in nature. New constellations of mobile streaming are not isolated but rather a part of a long lineage. And that lineage is driven by the desire to capture new times and spaces for entertainment, information, production, and monetization.

These are just some of the many developing aspects of mobile streaming. Mobile streaming encompasses such a range of platforms and practices, and it looks to remain a key modality of audio-visual interaction for the foreseeable future. We hope that by looking to the past development of mobile streaming, we can better understand our present moment. Ultimately, the trajectory of mobile personalization is not novel, even if what we're watching and listening to, and on what platform, is continually changing.

Looking ahead, the future of mobile streaming remains open-ended. It will continue to evolve alongside new devices and services but will probably remain the constant hum of life. It's neither an unalloyed blessing nor a curse; it is simply one of the many ways technologies are intertwined with culture and everyday life. Rather than feeling optimistic or pessimistic, perhaps the wisest stance is to stay curious and mindful of how the next stage unfolds. In the end, mobile streaming is here to stay and will keep reshaping the fabric of daily life—a flowing thread of contemporary existence, carrying our hopes and contradictions into whatever comes next.